



Neural compensation of residual distortion in camera calibration

KELEI WANG,¹ , JUNPENG XUE,^{1,*} FEIFEI HE,²
HONGYANG CHEN,³ AND RAN JIA¹

¹Key Laboratory of Advanced Spatial Mechanism and Intelligent Spacecraft, Ministry of Education, School of Aeronautics and Astronautics, Sichuan University, Sichuan 610065, China

²Sichuan Aerospace Changzheng Equipment Manufacturing Co., Ltd., Sichuan 610100, China

³Xichang Satellite Launch Center, Xichang 615000, China

*jpxue@scu.edu.cn

Abstract: Accurate camera calibration is crucial for achieving high-precision 3D measurements, particularly when using low-cost devices in long-range or complex depth environments where modeling distortion becomes significantly more challenging. Traditional calibration approaches often fail to capture focal-depth-dependent distortion due to their limited parametric structures, compromising reliability in demanding scenarios. To tackle these issues, we propose a two-step calibration process that combines geometric distortion modeling with learning-based refinement. In the first stage, we develop a Phased Distortion Attenuation Optimization (PDAO) algorithm, which gradually reduces distortion effects across varying depths. In the second stage, we introduce a neural network-based error compensation module (NN Model). It learns residual distortion patterns by taking image coordinates as input and producing undistorted coordinates based on the optimized intrinsic parameters. Experimental results demonstrate that our method significantly improves reprojection accuracy and robustness under wide-baseline and multiview settings. The proposed approach makes it particularly suitable for high-precision 3D vision applications in challenging or unconstrained environments.

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1. Introduction

Camera calibration is a fundamental technique in visual metrology and computer vision [1,2]. Lens distortion, a geometric warping caused by refraction and focusing of light through the lens, is typically decomposed into radial and tangential distortion. To ensure calibration accuracy, various geometric models have been developed. Zhang's planar calibration method requires observing a planar pattern from multiple orientations and models radial distortion through a closed-form solution followed by nonlinear refinement [3,4]. This method provides flexibility without expensive equipment, laying the groundwork for subsequent research. However, as optical designs become more complex and high-precision applications proliferate, traditional polynomial radial models face new challenges. They assume distortion is uniform across all fields of view and object distances [5]. While this holds for long-range measurements, it often fails in close-range or varying-depth scenarios, leading to reduced calibration accuracy.

Camera calibration spans fields including optical metrology, 3D reconstruction, augmented reality, and robotic vision [6], and its accuracy directly affects system performance. In these domains, precise distortion correction is a key step for ensuring measurement accuracy and image quality. As noted in the literature, camera calibration is critical in structured-light and projector-based 3D measurement systems [7]. In medical imaging, industrial inspection, and AR applications, high-precision calibration substantially improves positioning and reconstruction reliability. Researchers therefore seek more universal and refined distortion correction methods to meet the diverse demands of these applications. Madejski emphasized that lens distortion is a

common source of systematic error, and correcting it can dramatically reduce reprojection error [8].

Traditional distortion correction uses parameterized models that fit radial and tangential terms to images of a calibration pattern, typically assuming distortion is independent of object distance [9,10]. For example, these models work well for distant targets but incur larger errors in near-range or varying-depth scenarios [11–13]. Camera calibration for wide-angle and fisheye lenses has been widely studied. The model proposed by Kannala and Brandt [14] offers a solution for calibrating conventional, wide-angle, and fisheye lenses by employing a polar coordinate parameterization, effectively addressing the distortion characteristics of these lenses. Ricolfe-Viala [15] observed that for strong-distortion fisheye lenses, the distortion changes significantly with object distance or focus settings, a depth coupled effect poorly handled by a single fixed model.

Researchers have begun explicitly addressing focal-depth-dependent. Magill showed radial distortion depends on magnification but lacked an invertible distortion function for infinite focus [16]. Brown extended this to a practical focal-depth-dependent radial distortion model [17], and Fryer added a focal-depth-dependent tangential term [18]. These models are comprehensive but involve many parameters and complex calibration: lenses must be refocused for each depth during calibration, making the process time-consuming. For example, Sun proposed a model with both radial and tangential focal-depth-dependent, but it had over 16 unknown parameters to calibrate, hindering convergence [19].

More recently, deep learning approaches have emerged for camera calibration. These model-free methods use neural networks to implicitly learn the nonlinear mapping between image pixels and scene coordinates. Studies show that NNs offer stronger robustness to noise and distortion. Dou et al. systematically evaluated NN-based spatial calibration and found the NN model outperforms conventional pinhole and polynomial models in robustness and accuracy under various conditions [20]. Similarly, Hiroi et al. introduced the Neural Distortion Field (NDF), a fully-connected network representation of distortions in wide-FoV near-eye displays. With only a few training viewpoints, NDF achieved a median calibration error of 3.23 pixels and outperformed nonlinear polynomial fits, especially near the center of the view [21]. Cao et al. proposed a model-free stereo calibration for fisheye cameras: their NN directly learns the mapping between image coordinates and spatial points without pre-correcting images, yielding superior results for severe distortions [22]. In structured light 3D measurement, Chung et al. designed an NN that takes pixel position and object height as input, reducing the X/Y coordinate error by 60% compared to a conventional model [23]. These works demonstrate the flexibility of neural methods in capturing complex distortions, but also highlight a common challenge: the true undistorted pixel coordinates needed for supervision are not directly available in real scenes. Genovese proposed a single-image, model-free calibration using digital image correlation to obtain a dense distortion map [7]. Similarly, Chaudhry demonstrated that a ResNet-based model can predict the full set of camera intrinsics and Brown–Conrady distortion coefficients from one image, using a synthetic dataset for training [24]. While these methods offer novel capabilities, they present certain limitations. The former often requires specialized hardware, while the latter is fundamentally limited by the expressive power of the parametric Brown–Conrady model it predicts and faces a potential domain gap due to its reliance on synthetic data. Such parametric models often fail to capture complex, non-polynomial lens distortions.

To address these issues, this paper proposes a Z-axis coupled neural network model for distortion and a stage-wise Distortion Attenuation Optimization (PDAO) algorithm to significantly reduce reprojection error. Specifically, we use a neural network to implicitly learn the mapping between pixel coordinates and the distortion field at different depths, so that the model accounts for depth variations in 3D space. Simultaneously, we design a stage-wise optimization process by combining the initially estimated calibration coordinates with the network's outputs and

iteratively refining them, we progressively attenuate distortion effects. These objectives aim to improve the distortion model's accuracy and the consistency of the calibration results by leveraging depth information and iterative optimization.

The key problems are how to build a depth coupled distortion model and how to obtain accurate undistorted supervision targets. We analyze experimental observations confirming that a given pixel exhibits significantly different distortion direction and magnitude at different Z-depths. Based on this, we use a neural network as a mapping function. The inputs are the pixel coordinates and the corresponding Z-depth, and the output is the pixel's position in an ideal undistorted image. The neural network implicitly learns the 3D distortion field from ample samples, enabling the model to adapt to distortion changes at different object distances. Since true undistorted targets cannot be directly obtained in practice, we design an iterative PDAO algorithm. We use a conventional calibration to get initial intrinsic parameters and distortion, and compute an initial guess of undistorted pixel coordinates. This process is carried out in multiple stages, each progressively attenuating the effect of distortion on the coordinates, so that the estimated calibration points approach the true undistorted positions.

The rest of this paper is organized as follows. Section 2 reviews related work and analyzes focal-depth-dependent camera distortion; Section 3 introduces the stage-wise distortion attenuation optimization (PDAO) algorithm and Z-axis coupled neural network distortion model; Section 4 presents experimental results and analysis; Section 5 concludes the paper.

2. Analysis of visual imaging models and distortion factors

2.1. Ideal camera imaging model

The ideal camera imaging model is shown in Fig. 1 [25], where $O_w|\hat{X}_w\hat{Y}_w\hat{Z}_w$ is the world coordinate system, $O_c|\hat{X}_c\hat{Y}_c\hat{Z}_c$ is the camera coordinate system, $O_i|\hat{x}\hat{y}$ is the image plane coordinate system, and $O_p|\hat{u}\hat{v}$ is the pixel coordinate system.

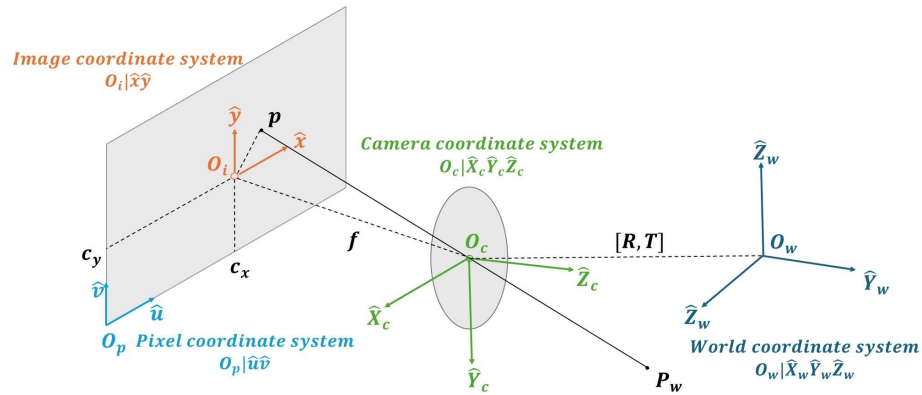


Fig. 1. Ideal Camera Imaging Model.

Point P in the world coordinate system has coordinates $(X_w, Y_w, Z_w)^T$ and its corresponding homogeneous coordinates are $(X_w, Y_w, Z_w, 1)^T$. The coordinates of point P in the camera coordinate system are $(X_c, Y_c, Z_c)^T$. The transformation from the world coordinate system to the

camera coordinate system is a rigid transformation, and the transformation formula is as follows.

$$\begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix} = \begin{bmatrix} R_{3 \times 3} & T_{3 \times 1} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} \quad (1)$$

where $R_{3 \times 3}$ is the rotation matrix and $T_{3 \times 1}$ is the translation matrix [26].

The coordinates of the image point p in the image plane coordinate system are $(x, y)^T$, and the coordinates in the image coordinate system are $(u, v)^T$. According to the principle of similar triangles, the relationship between the coordinates of point P in the camera coordinate system and the coordinates of its image point p in the image plane coordinate system is:

$$\begin{cases} x = f \frac{X_c}{Z_c} \\ y = f \frac{Y_c}{Z_c} \end{cases} \quad (2)$$

The relationship between the image coordinate system and the pixel coordinate system is:

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x & 0 \\ 0 & f_y & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = K \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad (3)$$

where (f_x, f_y) are the focal lengths, and (c_x, c_y) are the coordinates of the principal point in the pixel coordinate system. Therefore, the relationship between point p and point P in the ideal camera imaging model is:

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x & 0 \\ 0 & f_y & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} R_{3 \times 3} & T_{3 \times 1} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} \quad (4)$$

2.2. Camera distortion model

In reality, cameras use lens groups to achieve pinhole imaging, and manufacturing and assembly errors in the lenses introduce lens distortions to the imaging process [27]. The relationship between the distortion amount and the image coordinates is given by [28]:

$$\begin{bmatrix} x_d \\ y_d \end{bmatrix} = \begin{bmatrix} x + \Delta x \\ y + \Delta y \end{bmatrix} \quad (5)$$

where (x_d, y_d) are the true imaging coordinates of the target point in the camera imaging model's image coordinate system, (x, y) are the distorted coordinates, Δx and Δy are distortion amounts of the target point in two directions.

The distortions in the lens are typically described using polynomials [29] and are categorized into radial distortion, tangential distortion, and thin prism distortion. Generally, camera calibration

only considers radial and tangential distortions, and the distortion amounts are calculated as follows:

$$\begin{cases} \Delta x = x \cdot (k_1 \cdot r_c^2 + k_2 \cdot r_c^4 + k_3 \cdot r_c^6 + \dots) + 2p_1xy + p_2(r_c^2 + 2x^2) \\ \Delta y = y \cdot (k_1 \cdot r_c^2 + k_2 \cdot r_c^4 + k_3 \cdot r_c^6 + \dots) + 2p_2xy + p_1(r_c^2 + 2y^2) \\ r_c^2 = x^2 + y^2 \end{cases} \quad (6)$$

where k_1, k_2 and k_3 are the radial distortion coefficients, and p_1 and p_2 are the tangential distortion coefficients. Since the influence of high-order distortion on the results is almost negligible, this paper considers only k_1 and k_2 . After incorporating lens distortion, the coordinates of the image point become:

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K \begin{bmatrix} x_d \\ y_d \\ 1 \end{bmatrix} \quad (7)$$

2.3. Model solution

The solution of camera imaging model parameters is typically divided into two parts [30]. First, ignoring the nonlinear lens distortion, the intrinsic and extrinsic parameters of the camera are solved, which becomes a linear problem. The homography matrix H for each image can be obtained using singular value decomposition (SVD), and then it is decomposed into the corresponding intrinsic matrix K , rotation matrix R , and translation vector T , where $H = K[R, T]$. Subsequently, considering the nonlinear lens distortion, the objective function is constructed to minimize the reprojection error:

$$\min_{K, R, T, k_1, k_2, p_1, p_2, i=1, j=1} \sum_{i=1}^m \sum_{j=1}^n \|p_{ij} - Q(K, R_i, T_i, k_1, k_2, p_1, p_2, P_j)\|^2 \quad (8)$$

where m is the number of images, n is the number of feature points on each image, R_i and T_i are the extrinsic parameters of the i -th image, P_j is the world coordinate of the j -th feature point, p_{ij} is the detected feature point on the i -th image, and $Q(K, R_i, T_i, k_1, k_2, p_1, p_2, P_j)$ is the corresponding projected point of the camera model. Using the solution obtained in the first step as the initial value, the Levenberg-Marquardt optimization algorithm is employed to solve the parameters of the camera model [31].

2.4. Distortion model analysis and simulation

Figure 2 shows the checkerboard pattern used in the experiments, where the calibration board features squares with a length of 1 mm. The images depicted below were utilized for calibration purposes.

Given that the image size varies with height, additional planes at $Z = 0, \pm 10\text{mm}$, and $\pm 20\text{mm}$ were incorporated. The world coordinates and their corresponding image coordinates were substituted into the equation, and the transformation matrix was derived through the least squares method (LSM).

In theory, the computation of the coordinate transformation matrix should employ undistorted image coordinates. However, since these are unattainable directly, distorted image coordinates were utilized for estimation [32]. As distortion errors manifest in both the x and y directions, they were projected onto the xy plane for concurrent representation. Figure 3 illustrates the measurement errors under four distinct distortion models, with the rectangular frames in Figs. 2 and 3 representing the benchmark quadrilaterals employed for precision testing. The rectangular frame connects the four outermost corner points of the checkerboard: $(-7, 7), (7, 7), (7, -7)$ and

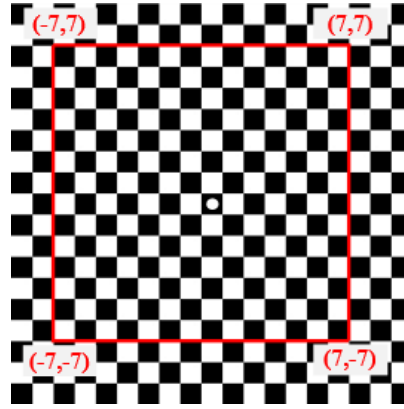


Fig. 2. Checkerboard pattern.

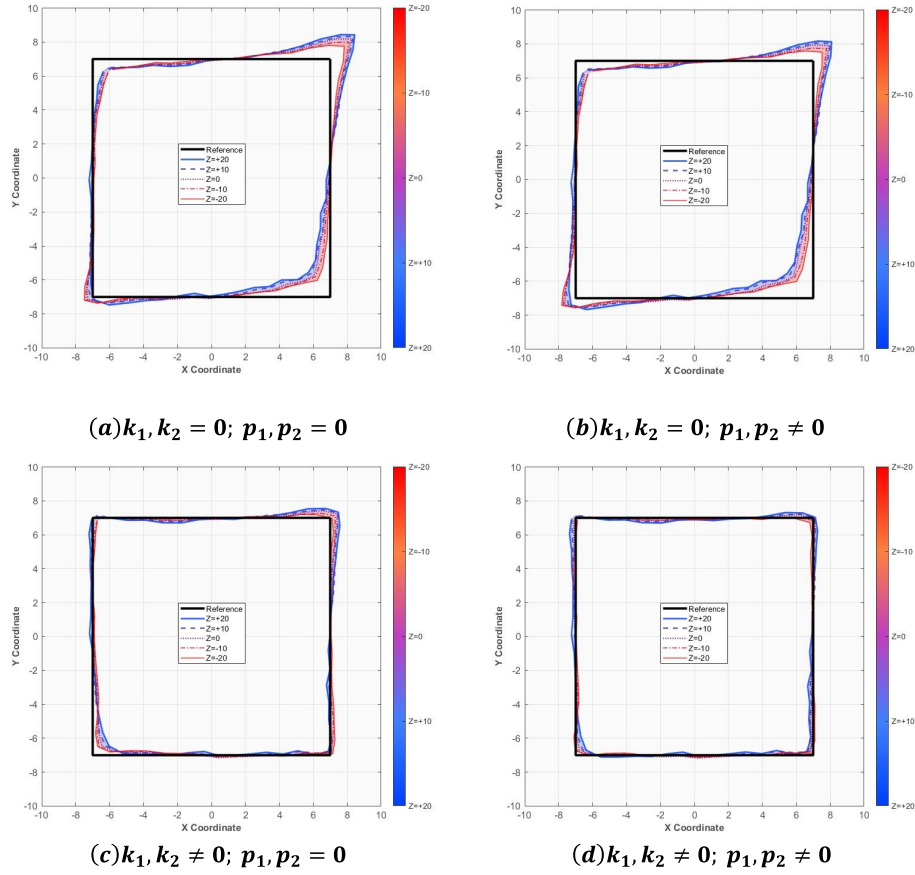


Fig. 3. Measurement errors under four distinct distortion models.

$(-7, -7)$ mm, encompassing a total of 56 reference points. The non-black lines in the figure depict the offset trajectories after distortion.

The x and y axes in the figure are denoted in millimeters (mm), indicating the positions of the test points, while the error is expressed in pixels, reflecting the displacement at each point.

Figure 3(a) presents the error distribution without any distortion correction. In this scenario, the curves are most dispersed, particularly at the four corners, where the contours of different planes deviate most significantly from the black reference line. Consequently, without any distortion correction, as the imaging plane radius increases, both radial and tangential distortions accumulate, resulting in the largest reprojection coordinate errors, which can reach several pixels. Figure 3(b) shows the error distribution when only tangential distortion correction is introduced, without radial distortion correction. The curve shape is similar to that without any distortion correction, with only slight changes in the offsets at the corners and edges, but the overall distribution remains highly dispersed. Since tangential distortion primarily corrects for image lateral stretching caused by lens misalignment and manufacturing defects, it has limited impact on the overall radial curvature. Therefore, compensating for tangential distortion alone does not significantly converge the measurement contours. Figure 3(c) displays the error distribution when only radial distortion correction is introduced, without tangential distortion correction. The contours of different planes have significantly converged towards the reference rectangle, with the substantial distortions at the corners largely eliminated. Radial distortion is the primary source of lens distortion, and compensating for it alone effectively corrects errors away from the optical center. Figure 3(d) illustrates the error distribution when both radial and tangential distortions are corrected. The curves are the most converged, fitting most closely with the black reference frame. With radial distortion already largely corrected, introducing tangential compensation further refines the remaining minor asymmetric residual errors (e.g., slight lens tilt or assembly deviations), thereby achieving the optimal in-plane positioning accuracy.

In Fig. 3(d), it is evident that at $Z = +20\text{mm}$, there is a noticeable protrusion, while at $Z = -20\text{mm}$, a reverse offset may occur. Moreover, the direction and magnitude of the offsets on different planes do not follow a simple pattern of equal-amplitude scaling or translation; instead, both the shape and direction vary with depth [4]. Although this phenomenon appears to contradict the pinhole camera principle, the pinhole model is an ideal distortionless imaging system. In practice, lens manufacturing quality significantly influences the complexity of the distortion model. For fixed-focal-length lenses, distortion parameters are theoretically fixed, but the aperture size determines the depth of field for clear imaging. At different positions within the depth of field, particularly at the blurred edges, the distortion parameters differ from those at the optimal focus point in the middle, limiting traditional single model approaches from accurately characterizing distortions across the entire measurement range. Therefore, in the subsequent steps, we will incorporate Z into the lens distortion model and utilize a neural network [33] to learn the influence of Z on distortion, further enhancing the accuracy of the calibration.

3. Phased distortion attenuation optimization and neural network modeling (PDAO-NN)

This chapter introduces the PDAO-NN method, which uses a neural network to address the residual distortion issues identified earlier. The PDAO procedure provides more accurate data, helping the network to effectively learn and correct these distortions. The overall technical route is shown in Fig. 4.

The comparison of the conventional forward projection model and the proposed correction framework is shown in Fig. 5. Zhang's method is the algorithm used to find the parameters of forward projection.

3.1. Calibration foundation and initial model establishment

In camera calibration, there are two main paradigms: pattern-based and pattern-free. Pattern-free methods, represented by recent state-of-the-art works like GeoCalib [34] (which uses geometric cues from a single image) and Neural Lens Modeling [29] (which learns from multi-view images of a general scene), offer incredible flexibility and are powerful for in-the-wild applications or

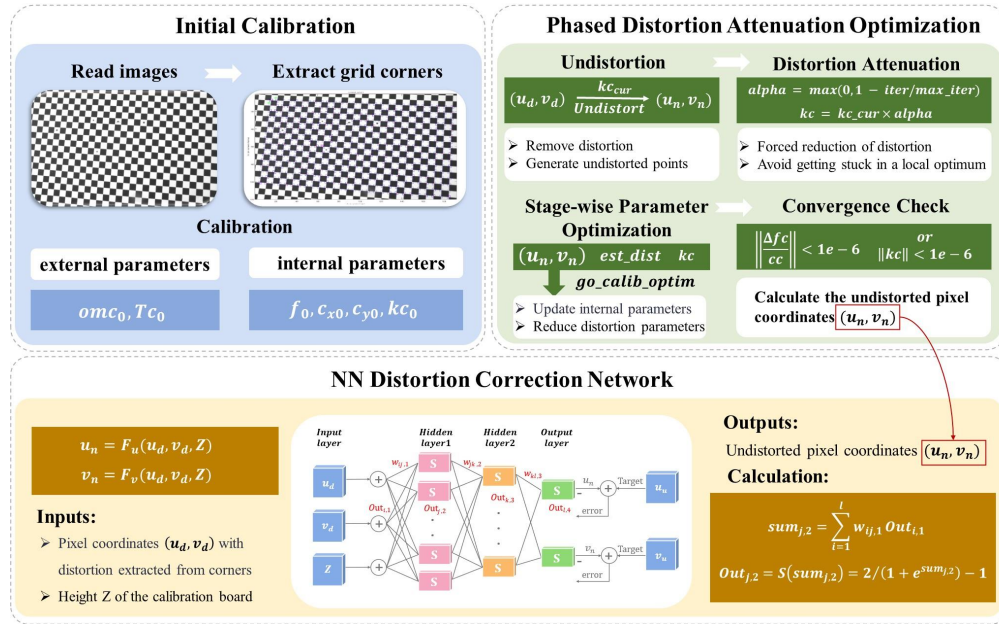


Fig. 4. Technical route.

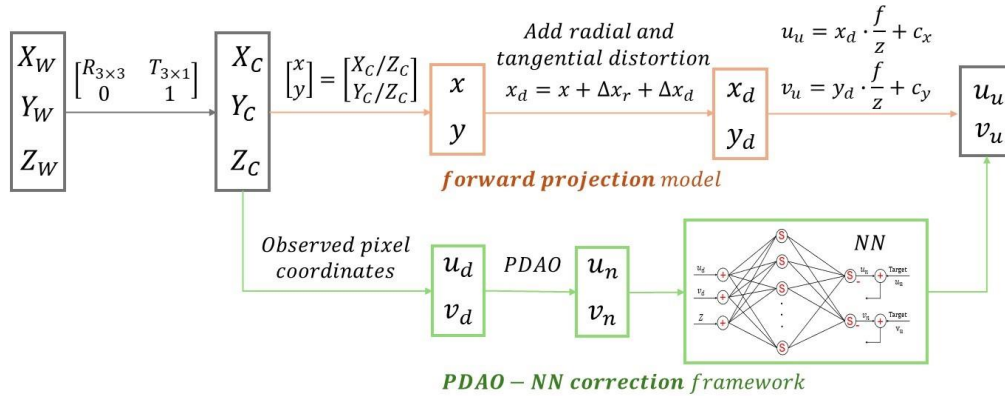


Fig. 5. Conceptual Comparison of the Conventional Forward Projection Model and the Proposed Correction Framework.

graphics. However, their reliance on less constrained natural features means they often cannot achieve the level of accuracy required for high-precision metrology. For such tasks, pattern-based methods like Zhang's planar calibration [3] remain the standard due to the strong geometric ground truth provided by the known pattern. Therefore, this study is positioned within the high-precision, pattern-based paradigm. We first use Zhang's method to provide a reliable initial calibration, and our primary contribution is a novel framework to learn and compensate for the complex residual errors, such as focal-depth-dependent distortion, that are not captured by the initial model. This method utilizes images of a planar chessboard with known dimensions, captured from multiple different viewpoints. By extracting corner features, it calculates the mapping relationship between the world coordinate system and the image plane in each image and jointly solves for the camera's intrinsic and extrinsic parameters.

After the initial calibration is completed, the obtained parameters include: the focal length vector of the camera $\mathbf{f} = [f_x, f_y]$, the principal point coordinates $\mathbf{c} = (c_x, c_y)$, the radial and tangential distortion parameters $\mathbf{d} = [k_1, k_2, p_1, p_2, k_3]$, the rotation and translation vectors omc_i and T_i for each image, as well as the pixel coordinates (u_i, v_i) of the corner points in each image and their corresponding chessboard world coordinates (X_i, Y_i, Z_i) .

3.2. Generating undistorted ground truths via phased distortion attenuation optimization (PDAO)

A fundamental challenge for training any learning-based distortion correction model is the absence of ground truth data. In a real-world setting, the ideal, perfectly undistorted image coordinates corresponding to observed points are unknown and cannot be directly measured. A naive approach might use the undistorted coordinates calculated from a single run of a traditional method like Zhang's as training targets. However, these targets would still be contaminated by the residual errors of the parametric model itself, leading the neural network to simply learn and replicate an imperfect model.

To solve this "ground truth generation" problem, we propose the phased distortion attenuation optimization (PDAO) algorithm. The primary and sole function of PDAO is to computationally generate a set of high-fidelity, undistorted target coordinates that can serve as reliable supervision labels for training our neural network.

It is critical to note that PDAO is not a data augmentation technique. Data augmentation aims to increase the diversity and quantity of training inputs. In contrast, PDAO is a pre-processing step focused entirely on refining the training outputs, producing the most accurate possible estimation of the ideal, zero-distortion state.

The core principle of PDAO is to start with an initial camera calibration and iteratively re-estimate the camera's intrinsic parameters while systematically forcing the distortion coefficients $\mathbf{d}$ to zero. This process effectively isolates the pure geometric projection from the distortion effects, yielding a final set of undistorted coordinates that are more accurate than those from a one-step traditional method. The optimization process is a double layer loop. The outer layer iterates to update the intrinsic parameters, while the inner layer corrects the observed image points for distortion to obtain undistorted estimates under the current parameters. The specific process is shown in Fig. 6.

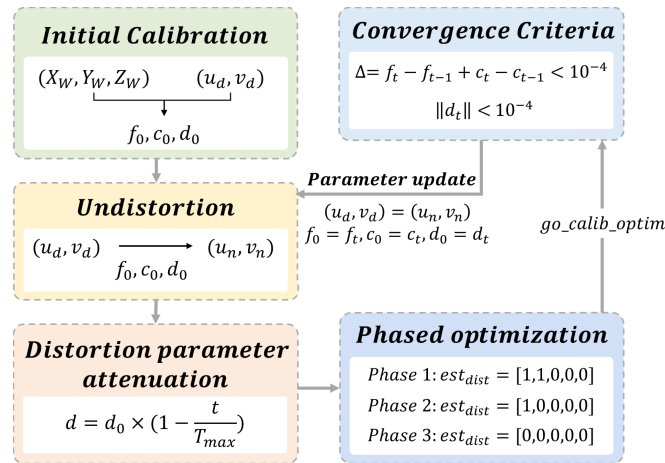


Fig. 6. Algorithm Flow.

Step 1: Undistortion. In each iteration, the original distorted corner points detected are converted to undistorted points under the current intrinsic parameters. This step is achieved through iterative backprojection:

$$u_{i,t+1} = \text{undistort}(u_i^{\text{obs}}; \mathbf{f}, \mathbf{c}, \mathbf{d}_t) \quad (9)$$

where $\mathbf{d}_t$ represents the distortion parameters at the current iteration, and the back-projection process stops when the change in the image point position is less than a threshold of 10^{-3} or when the maximum number of iterations $N_{\max} = 50$ is reached.

Step 2: Dynamic attenuation of distortion factors. To stabilize the optimization process, PDAO applies an attenuation factor α to the distortion parameters in each iteration, gradually reducing them to zero:

$$\mathbf{d}_t = \alpha_t \cdot \mathbf{d}_{(t-1)}, \quad \alpha_t = \max \left(0, 1 - \frac{t}{T_{\max}} \right) \quad (10)$$

where $T_{\max}$ is the total number of iterations (set to 50 in this paper). This strategy ensures that distortions have a guiding effect in the early stages of optimization and gradually exit the modeling process, transitioning smoothly from "distorted modeling" to "zero-distortion modeling."

Step 3: Phased distortion model setting. To avoid estimating multiple highly coupled distortion parameters simultaneously, PDAO further defines a phased activation strategy est_{dist} to control which distortion parameters are activated. Specifically, Phase 1 (1-10 iterations): Estimate both k_1 and k_2 , with $est_{dist} = [1, 1, 0, 0, 0]$; Phase 2 (11-20 iterations): Estimate only k_2 , with $est_{dist} = [1, 0, 0, 0, 0]$; Phase 3 (21 iterations and beyond): No distortion parameters are estimated, maintaining $est_{dist} = [0, 0, 0, 0, 0]$;

Step 4: Parameter optimization. After undistortion and distortion model setting, the parameters are optimized using the "go calib optim" function, updating both intrinsic and extrinsic parameters.

Step 5: Convergence judgment. After each update, two indicators are calculated to determine convergence. The convergence criteria are:

$$\Delta = f_t - f_{t-1} + c_t - c_{t-1} \quad (11)$$

If $\Delta < 10^{-4}$, it is considered that the main parameters have converged. If the distortion has already approached zero, it can also be considered that the optimization task is complete.

3.3. Neural network (NN) modeling

Based on the previous analysis, distortions introduced by lens groups and manufacturing variations lead to focal-depth-dependent effects. Therefore, lens distortion varies with the position of the image plane and also changes with the distance to the object. By incorporating the object's height Z , the following new lens compensation model can be proposed.

$$u_n = F_u(u_d, v_d, Z) \quad (12)$$

$$v_n = F_v(u_d, v_d, Z) \quad (13)$$

The proposed modeling equation uses the distorted image coordinates (u_d, v_d) and height Z as inputs to find the undistorted image coordinates (u_n, v_n) . However, obtaining a new functional form for lens distortion by adding the Z variable is not straightforward. Even if a new modeling function is scarcely found, obtaining relevant coefficients is much more challenging. Therefore, we propose a neural network (NN) model. The NN model is used to represent multi-input/output functions and consists of multiple layers, including the input and output layers. Thus, given the number of nodes in the input and output layers and each hidden layer, a unique NN model is defined. Hidden layers are those that do not include the input and output layers. For simple

models, one hidden layer is used, while more complex models may employ two or three hidden layers.

Figure 7 shows an NN model with one hidden layer when the inputs are u_d, v_d and Z . The "+" node indicates that the inputs on the left are simply added to the right, but the "S" node performs very complex calculations. That is, at the j -th node of the hidden layer (second layer), the input values are multiplied by weights and summed, and then this value is transformed through a squash function to obtain an output value between -1 and $+1$. These calculations can be expressed as follows:

$$sum_{j,2} = \sum_{i=1}^I w_{ij,1} Out_{i,1} \quad (14)$$

$$Out_{j,2} = S(sum_{j,2}) = 2/(1 + e^{-sum_{j,2}}) - 1 \quad (15)$$

where I is the number of inputs, $w_{ij,1}$ is the weight factor connecting the i -th output of the input layer and the j -th node of the hidden layer, $Out_{i,1}$ is the output of the i -th node of the input layer, and $Out_{j,2}$ is the output of the j -th node of the hidden layer. The same calculation is repeated in the output layer (third layer). Since the output of the NN model is initially obtained randomly based on the given weight factors, the initial error compared to the desired output is large. Therefore, it is necessary to adjust the weights of the NN model through learning to minimize the error. In the NN model, the backpropagation algorithm is used for learning [35].

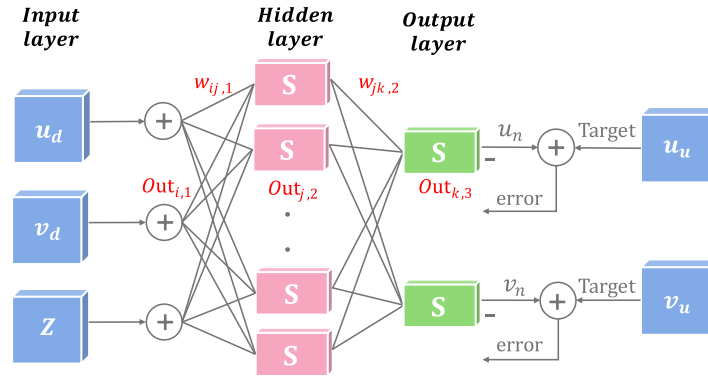


Fig. 7. One hidden layer model.

Figure 8 shows an NN model with two hidden layers when the inputs are u, v , and Z . To determine which network structure is most suitable for the experimental conditions in this paper, we compared the performance of different hidden layer structures of neural networks in terms of pixel coordinate reprojection error under the same training and validation datasets. The network structures "16–8" to "32–16–8" represent the number of nodes in each layer of the two-layer and three-layer hidden layers, respectively. By calculating the standard deviation (STD) and root mean square error (RMSE) of the errors in the u, v directions and the total error for each structure, the results in Table 1 were obtained.

As shown in the table, the "20–10" network achieves the minimum values in the two key metrics of total STD and total RMSE. Compared with networks of greater width or more layers, the "20–10" structure has a total of approximately 390 parameters, which ensures sufficient fitting capability while reducing training time. Therefore, the "20–10" structure is ultimately selected as the hidden layer architecture of the neural network distortion compensation model in this study, to balance model efficiency and robustness while maintaining high accuracy.

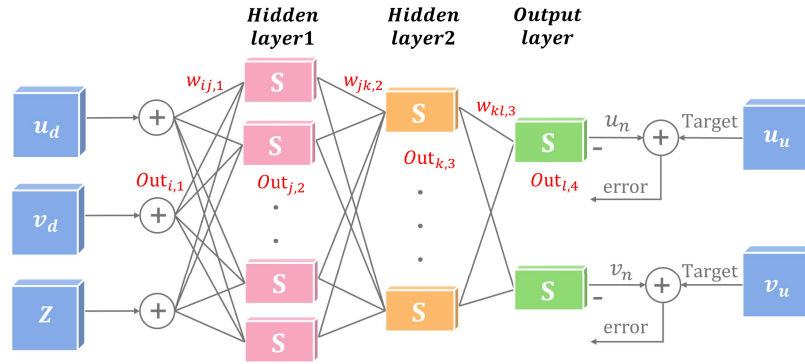


Fig. 8. Two hidden layer model.

Table 1. Error metrics under different network structures

NN structure	STD			RMSE		
	u-direction	v-direction	total	u-direction	v-direction	total
16-8	0.0735	0.0639	0.0596	0.0735	0.0639	0.0974
32-16	0.0611	0.0750	0.0648	0.0750	0.0648	0.0991
20-10	0.0653	0.0577	0.0533	0.0653	0.0577	0.0872
40-20	0.0711	0.0616	0.0575	0.0711	0.0616	0.0941
20-10-5	0.0703	0.0611	0.0569	0.0703	0.0611	0.0931
32-16-8	0.0726	0.0628	0.0593	0.0726	0.0628	0.0960

4. Experiments and results

4.1. Hardware

The hardware configuration is shown in Fig. 9. We utilized an industrial camera (IDS UI3370CP-MGL, 1920 x 1200 pixels [36]) with a set of interchangeable prime lenses for multiple sets of calibration board captures. The focal lengths of the lenses used included 8mm, 12mm, 16mm, 35mm, and 50mm. Additionally, a low-cost camera was employed. The device was a generic USB board camera (model GXIVISION-S2M03, 1920 x 1080 pixels) equipped with a 1/2.8-inch CMOS sensor (2.9 μm pixel size). The precision displacement platform (Daheng Optics GCD-203300M [37]), which has a travel range of 300 mm and a positioning precision of ± 0.002 mm, and its accompanying controller (GCD030401M) were both sourced from Daheng Optoelectronics, used to provide accurate Z-values.

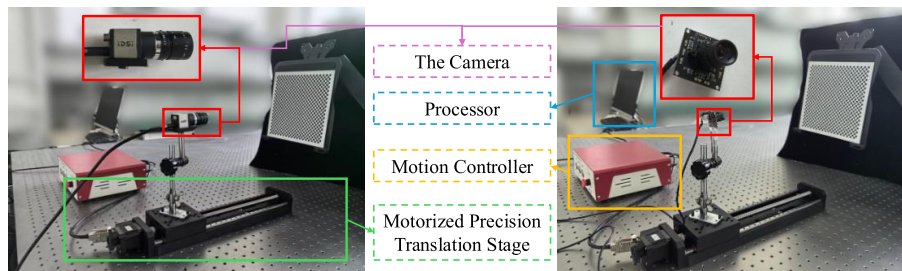


Fig. 9. Hardware.

4.2. Implementation details

Dataset Preparation: For each of the five fixed-focal-length industrial camera lenses, the dataset was constructed by capturing images at 5 distinct Z-axis positions. At each of these Z-positions, a set of 25 images was captured to ensure a robust calibration. These 25 distinct poses were generated through a combination of automated translation and manual orientation changes. The precision displacement platform was used to systematically step the calibration pattern through different positions across the camera's field of view. At each of these positions, the pattern was then manually tilted to several different orientations. This combined process yielded a well-distributed set of viewpoints. After completing the 25 poses, the platform was used to move the pattern to the next Z-axis position. The Z-positions ranged from 0 to +20 mm with an increment of 5 mm. The full dataset for each lens was then randomly split into 80% for training, 10% for validation, and 10% for testing.

Network Training Hyperparameters: The neural network was implemented and trained using the MATLAB Deep Learning Toolbox. The key hyperparameters and settings are summarized in Table 2. The Bayesian Regularization (trainbr) algorithm was chosen as it is a robust method for regression on smaller datasets. It is based on Levenberg-Marquardt optimization and automatically handles regularization to prevent overfitting. Notably, this algorithm does not require a manually set learning rate.

Table 2. Neural Network Training Hyperparameters

Parameter	Value / Method
NN Toolbox	MATLAB Deep Learning Toolbox
Training Algorithm	Bayesian Regularization (<code>trainbr</code>)
Optimizer	Levenberg-Marquardt
Performance Function	Mean Squared Error (<code>mse</code>)
Activation (Hidden Layers)	Hyperbolic Tangent Sigmoid (<code>tansig</code>)
Activation (Output Layer)	Linear (<code>purelin</code>)
Max Epochs	10000
Learning Rate	Not applicable for <code>trainbr</code>
Regularization	Handled automatically by Bayesian Regularization
Data Division	Random (<code>dividerand</code>)
Weight Initialization	Nguyen-Widrow
Data Normalization	Normalized to range [-1, 1]

4.3. Results of camera calibration

The partially captured calibration images are shown in Fig. 10.

The calibration parameters obtained using Zhang's method and the optimized parameters are shown in Table 3.

Table 3. Comparison of parameters from different methods

Method	f_x	f_y	c_1	c_2	k_1	k_2	p_1	p_2
Zhang	1717.65	1717.98	970.16	598.02	-0.1042	0.18783	-0.0002	0.00005
PDAO	1726.37	1726.48	969.22	595.94	0	0	0	0

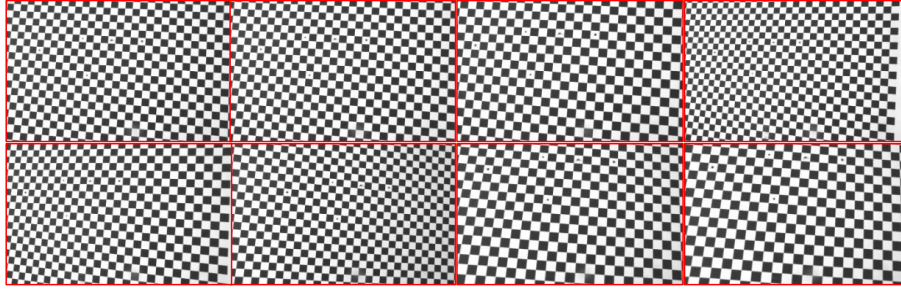


Fig. 10. Calibration images.

4.4. Comparison of distortion compensation model effects at different focal lengths

In this section, we compare the classic Zhang's polynomial model (hereinafter referred to as "Zhang") with the deep-coupled neural network model proposed in this paper (hereinafter referred to as "NN") in terms of reprojection error under various focal lengths. Table 4 summarizes the pixel-level standard deviation (STD) and root mean square error (RMSE) of the two models under different focal lengths for industrial cameras and ordinary low-cost cameras, as well as the error differences between them under the same focal length conditions.

Table 4. Calibration accuracy for different camera types and focal lengths

Camera Type	Focal Length	Model	STD	RMSE	Δ STD	Δ RMSE
Industrial Camera	8mm	Zhang	0.1482	0.2298	0.0918	0.1390
		NN	0.0564	0.0908		
	12mm	Zhang	0.0974	0.1656	0.0511	0.0847
		NN	0.0463	0.0809		
	16mm	Zhang	0.0571	0.1024	0.0130	0.0299
		NN	0.0441	0.0795		
	35mm	Zhang	0.0488	0.0867	0.0049	0.0084
		NN	0.0411	0.0750		
	50mm	Zhang	0.0327	0.0660	0.0019	0.0055
		NN	0.0308	0.0605		
Low-cost Camera		Zhang	0.3319	0.4248	0.1603	0.1688
		NN	0.1716	0.2560		

As shown in the table, the NN model outperforms Zhang's model in all tested focal lengths. At the shortest focal length, the STD is reduced from 0.1482 to 0.0564, and the RMSE is reduced from 0.2298 to 0.0908, with an error improvement of over 60%. Even at the longest focal length, the improvement in STD and RMSE remains around 5%. The trends of STD and RMSE with focal length for are shown in Fig. 11. As the object distance increases from 8 mm to 50 mm, the distortion of Zhang's model itself decreases, and the absolute improvement (Δ STD, Δ RMSE) of the NN model also decreases accordingly. However, even in the area with the weakest distortion in the far view, a small but stable accuracy improvement can still be obtained.

Figure 12 compares the median and interquartile range (IQR) of reprojection error across different focal lengths for both Zhang's calibration method and the proposed NN-based model. In Fig. 12(a), corresponding to Zhang's method, the median error decreases as the focal length increases from 8 mm to 50 mm, but the IQR remains relatively large, especially at shorter focal lengths, indicating greater variability and less stable calibration results. In contrast, Fig. 12(b),

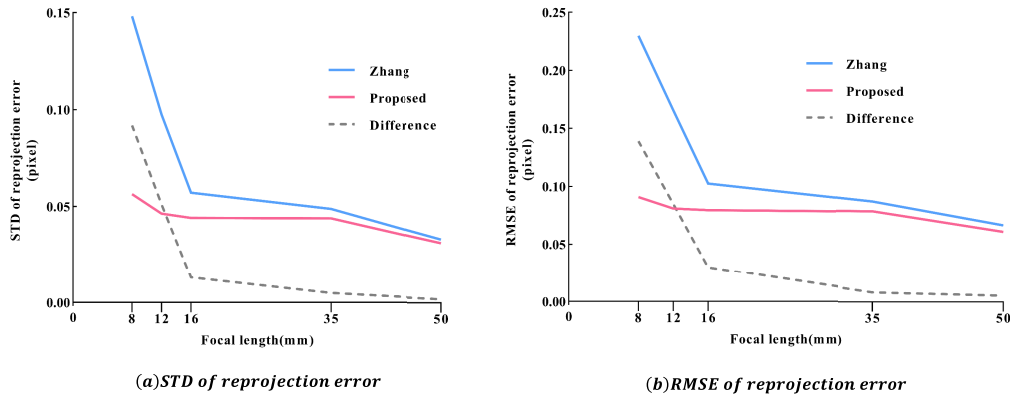


Fig. 11. The trends of STD and RMSE with focal length.

which shows the results of the proposed model, demonstrates consistently lower median errors and significantly smaller IQRs across all focal lengths. These results highlight that the proposed method not only reduces the average reprojection error but also provides more stable and consistent calibration performance under different imaging conditions.

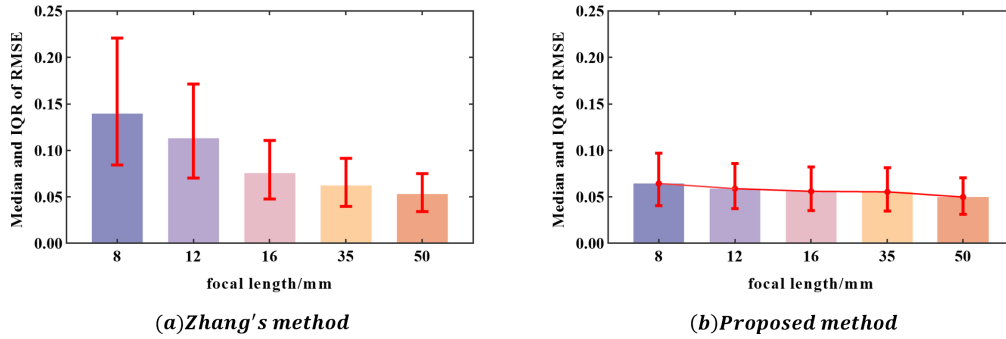


Fig. 12. The trends of the median and interquartile range (IQR) of reprojection error.

In the case of ordinary low-cost cameras, the STD and RMSE of Zhang's model are as high as 0.3319 pixels and 0.4248 pixels, respectively, indicating that the lens distortion is the most severe under this condition. The NN model reduces them to 0.1716 pixels and 0.2560 pixels, respectively, with an improvement of 40%–50%, which fully proves that the method has a significant calibration optimization effect on low-pixel cameras and is of great significance for the calibration optimization of low-pixel cameras.

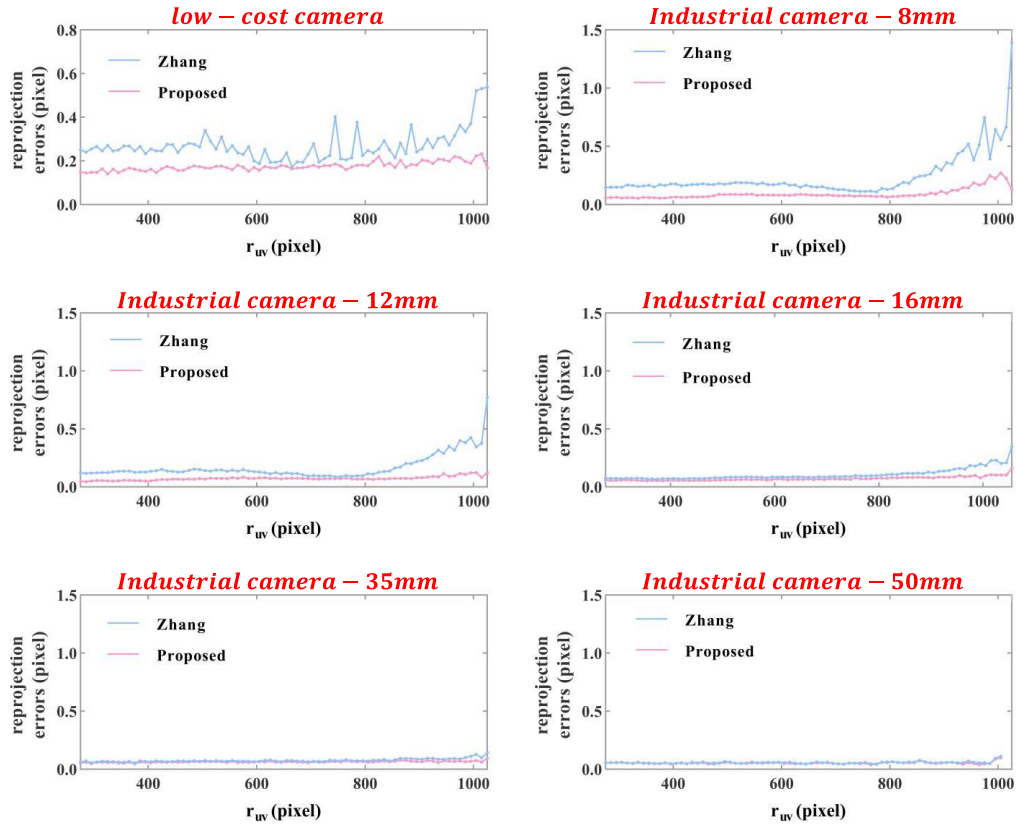
To further evaluate the consistency and repeatability of our calibration process, we conducted six independent calibration experiments under the same conditions with an 8 mm focal length. Table 5 summarizes the reprojection errors in terms of standard deviation (STD) and root mean square error (RMSE) for each run. The results show very small variations across the six experiments. This demonstrates the robustness and repeatability of the proposed calibration framework.

4.5. Analysis of reprojection error distribution at different focal lengths

Figure 13 shows the distribution trend of reprojection error with the distance from the image plane to the principal point under different focal lengths for different cameras. Under any focal length

Table 5. Repeatability results of six calibration runs at focal length 8 mm

Run	STD			RMSE		
	u-direction	v-direction	total	u-direction	v-direction	total
1	0.0727	0.0652	0.0600	0.0727	0.0652	0.0977
2	0.0765	0.0694	0.0623	0.0765	0.0694	0.1033
3	0.0719	0.0624	0.0590	0.0719	0.0624	0.0952
4	0.0698	0.0613	0.0568	0.0698	0.0613	0.0929
5	0.0702	0.0614	0.0569	0.0702	0.0614	0.0933
6	0.0706	0.0619	0.0576	0.0706	0.0619	0.0939

**Fig. 13.** Reprojection error with the distance to the principal point.

condition, the error of the NN model is always lower than that of Zhang's polynomial model. The three near views (8 mm, 12 mm, 16 mm) show a highly consistent change characteristic - when the radial distance is less than about 800 px (near the principal point), the error curve is smooth and the value is small; when the radial distance exceeds 800 px, the curve rises sharply, and the error increases significantly with increasing distance from the principal point. However, as the focal length increases from 8 mm to 16 mm, the overall maximum error value gradually decreases, indicating that the distortion in the far view itself weakens with increasing object distance. Under medium and long views (35 mm, 50 mm), since the overall distortion is relatively light, the error curve still rises slightly with distance, but the change is much flatter, and the trends of the two almost coincide, with the maximum error controlled within 0.1 mm. In the

case of ordinary cameras, the error distribution is the most flat, and a relatively stable calibration accuracy can be obtained.

Figure 14 shows the distribution of reprojection error of Zhang's method and the NN model under different conditions. The results show that under any condition, the reprojection error of the NN model is always lower than that of Zhang's polynomial model. The error gap between the two models remains consistent across different datasets, and no case of error reversal or abnormal increase was found under any condition. This indicates that by introducing the object distance, the NN model not only achieves excellent performance in ideal calibration environments but also has robustness against various complex acquisition conditions and can stably provide smaller reprojection errors than traditional algorithms.

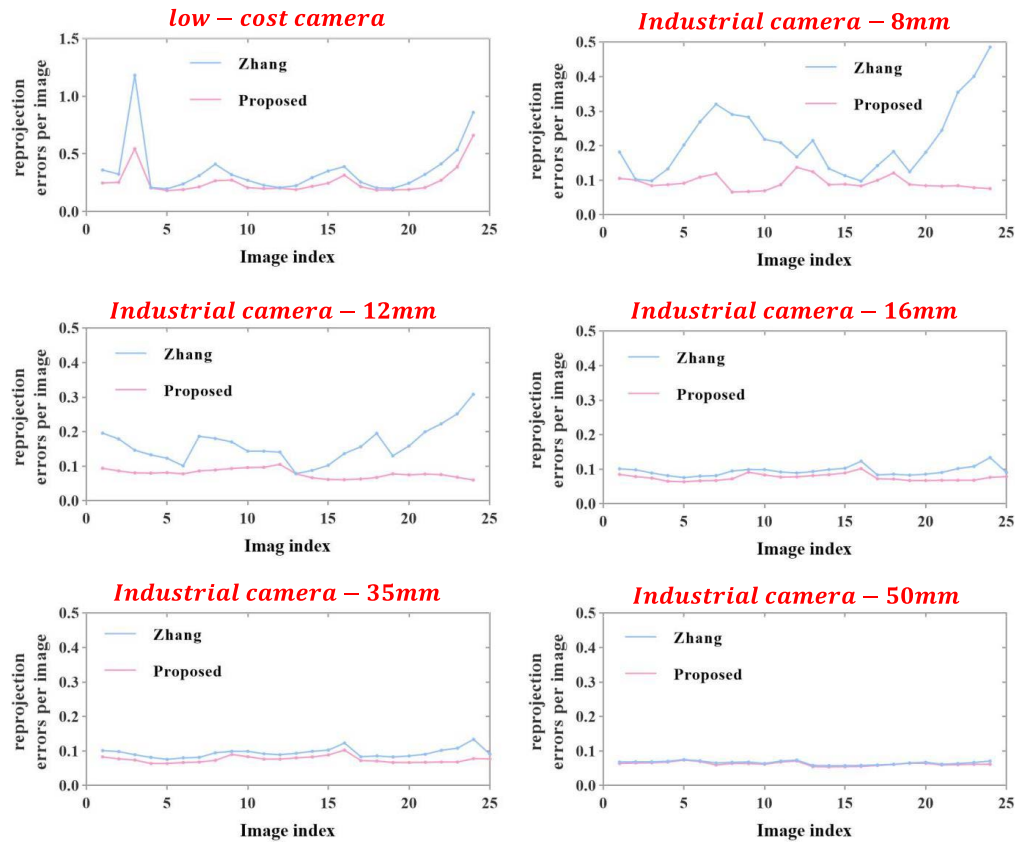


Fig. 14. Reprojection error of Zhang's method and the NN model under different images.

4.6. Visual comparison analysis of reprojection error

Figure 15 shows the reprojection error vectors of the entire calibration board feature points under Zhang's method and the NN method for the 8 mm industrial camera and the ordinary low-cost camera. Each circle marks the position of the original detected image corner point, each cross represents the position reprojectioned through the projection matrix or NN model, and the arrow points from the circle to the cross. The length of the arrow is magnified 300 times to highlight the error. It can be seen that the overall error level is reduced. Under both industrial and low-cost cameras, the arrows under the NN model are significantly shortened, indicating a significant reduction in average reprojection error. The edge error improvement is the most obvious. Under

Zhang's method, the arrows at the edge of the board are particularly long and the directions are inconsistent, while under the NN model, these edge arrows are much shorter, showing effective compensation for distortion. It can also be observed that a few vectors appear much larger than their neighbors and remain visible in both methods. These anomalies are caused by local inaccuracies in the feature point extraction process, which can be affected by external factors such as lighting conditions.

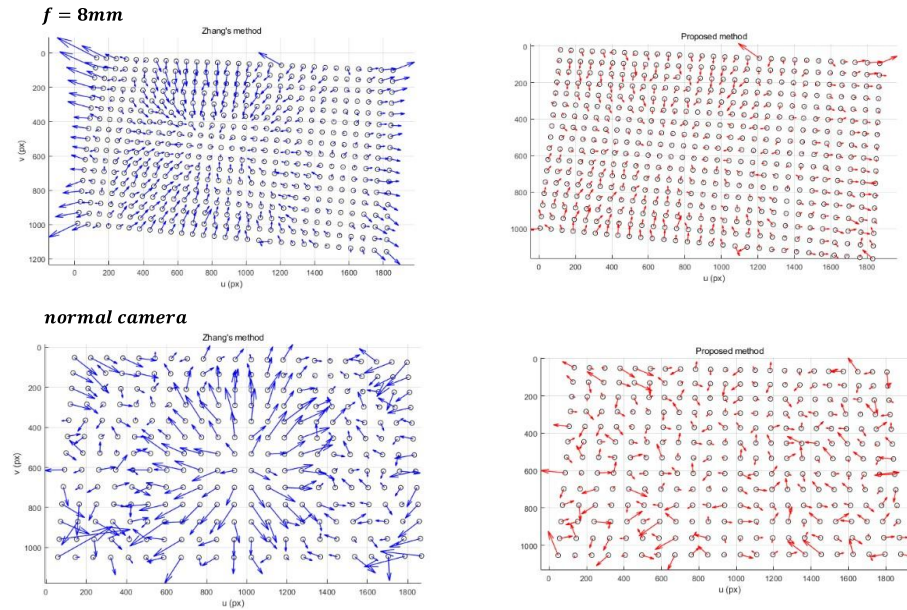


Fig. 15. Reprojection error vectors.

Figure 16 shows the scatter distribution of reprojection error in the plane coordinates for the 8 mm industrial camera and the ordinary low-cost camera, and a circle with a radius corresponding to twice the standard deviation is drawn with σ^2 as the center to intuitively show the degree of error concentration. The scatter cloud generated by the NN model is much more concentrated than that of Zhang's method, with most points falling within the σ^2 circle. Under Zhang's method, the scatter cloud has a wider distribution. For both cameras, the σ^2 value of the NN method is lower than that of Zhang's method, which quantitatively reflects the improvement in the tightness of the error distribution. The NN model not only reduces the average error at each point but also significantly reduces the fluctuation range of the error, improving the consistency and robustness of calibration.

5. Conclusion

This paper presents a novel calibration framework integrating phased distortion attenuation optimization (PDAO) and a depth-coupled neural network (NN) to address the limitations of traditional methods in edge distortion compensation and robustness under low-quality imaging. PDAO generates high-precision undistorted supervision signals through phased parameter attenuation and activation strategies, overcoming a key limitation in deep learning-based calibration that the reliance on imperfect ground truth coordinates generated by conventional models. The NN model pioneers depth information as an input variable, learning an end-to-end 3D nonlinear mapping from distorted to ideal coordinates, thereby enhancing adaptability to

strong distortions and wide fields of view. Systematic experiments on industrial cameras and low-cost devices demonstrate superior performance in standard deviation (STD) and root mean square error (RMSE): over 60% error reduction under strong distortion (8mm), 5%–8% accuracy gain for weak distortion, and 40%–50% error decrease in consumer-grade scenarios. This framework combines model-driven interpretability with data-driven generalization, establishing a new paradigm for high-precision calibration in complex imaging systems.

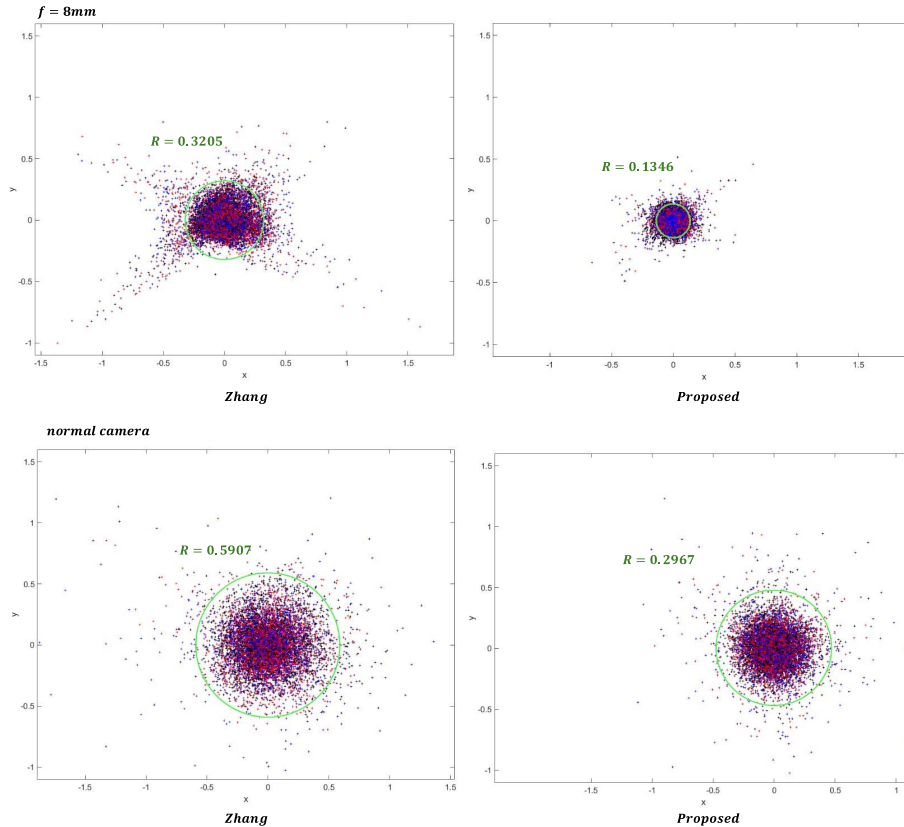


Fig. 16. The scatter distribution of reprojection error in the plane coordinates.

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